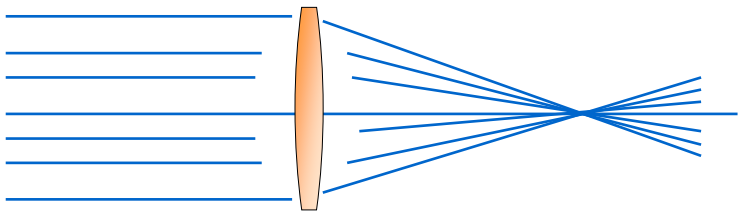
PHYC10009 Foundations of Physics

Imaging

17



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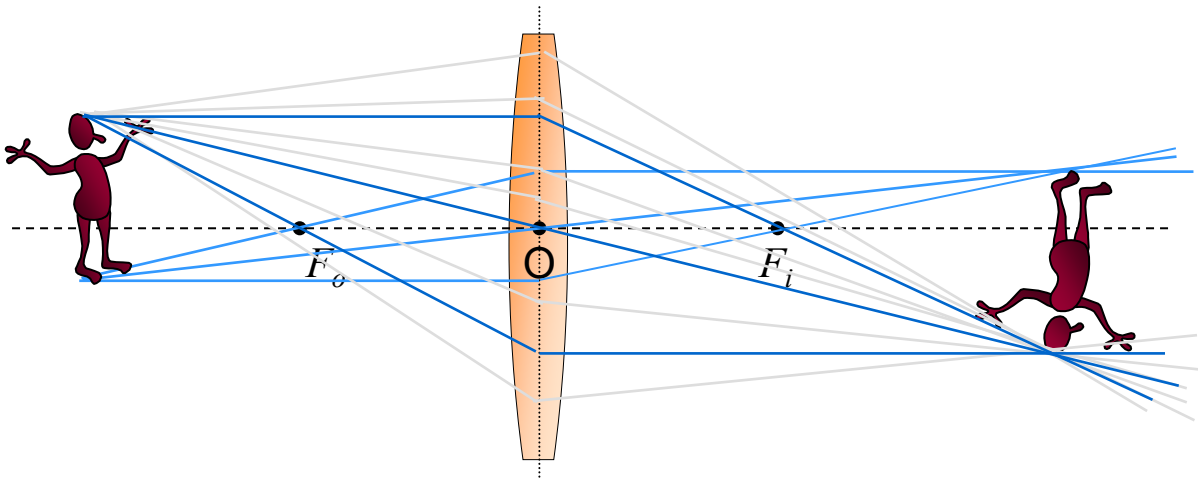
RAY TRACING: IMAGE FORMATION

Real images and ray tracing

- Real objects are not point sources
- Describe in terms of superposition of many points
- Trace many rays from many points… tedious
- Choose a few critical points and critical rays



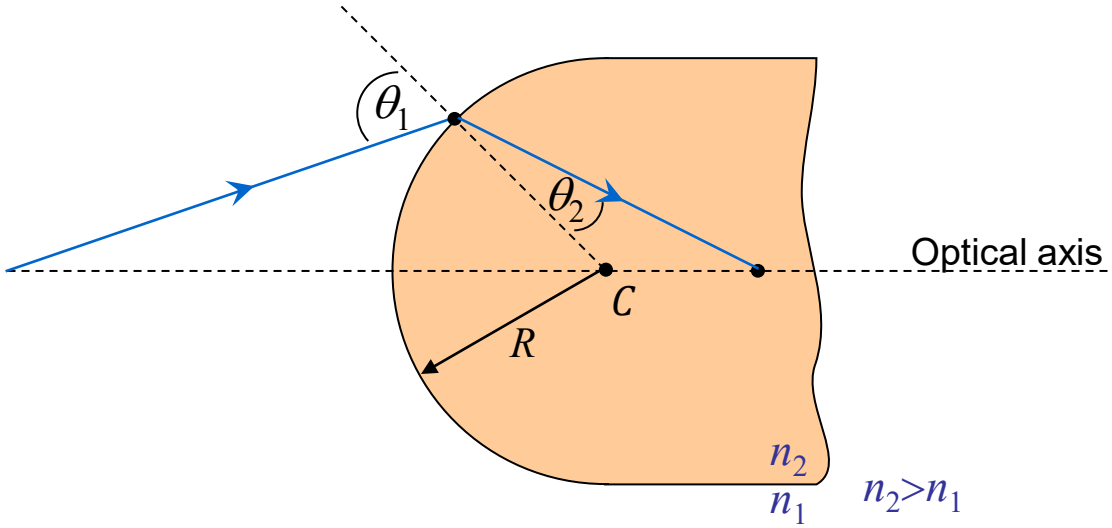
- For each point, draw three "special" rays:
  1. Through optic centre O
  2. Parallel to optic axis, then through far focal point F_i
  3. Through near focal point F_o then parallel to optic axis
- Actually only two of the three needed
- Transverse distance y_i, y_o are positive above the axis; note image inversion

NOTE: Object is not at infinite distance; rays not parallel, so $y = -f\theta$ must be used with care!

THIN LENS EQUATION

Spherical lenses: Snell's law

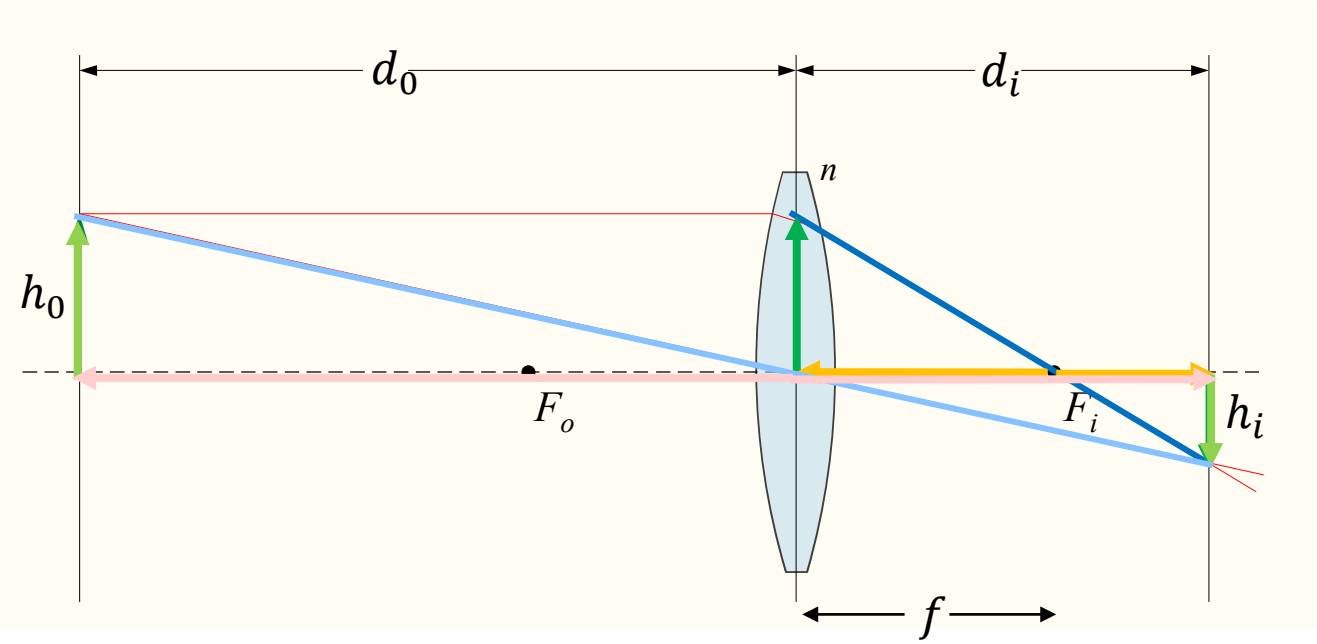
- How does a lens focus light?
- It's just Snell's law and trigonometry.



- Tricky … but there is a shortcut!

Thin lens

Imaging by a thin lens



Similar triangles: $\frac{h_i}{h_o} = \frac{d_i - f}{f}$ and $\frac{h_i}{h_o} = \frac{d_i}{d_o}$ $\Rightarrow \frac{d_i - f}{f} = \frac{d_i}{d_o} \Rightarrow \frac{d_i}{f} - 1 = \frac{d_i}{d_o}$ Divide by d_i

Thus $\frac{1}{f} - \frac{1}{d_i} = \frac{1}{d_0}$ or $\boxed{\frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}}$

Sign conventions

d_0	⇦ + if left of lens
d_i	⇨ + if right of lens
y, y'	⇧ + if above optical axis
f	+ if converging, – if diverging

Example: Thin lens equation

- $f = 100\text{mm}$ object at 600mm; where is image?

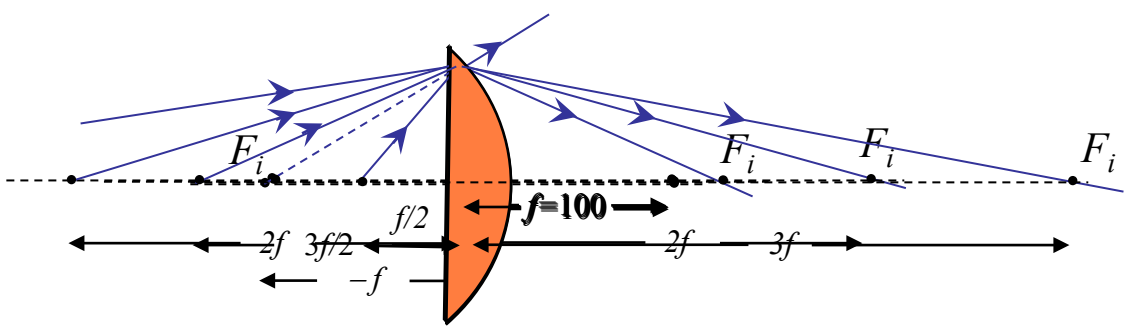
$$\boxed{\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}} \quad \Rightarrow \frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{d_o}{f d_o} - \frac{f}{f d_o} = \frac{d_o - f}{f d_o} \Rightarrow d_i = \frac{f d_o}{d_o - f} \quad d_i = \frac{d_o f}{d_o - f} = \frac{(600)(100)}{600 - 100} = 120\text{mm}$$

- Similarly,

$d_o = 200\text{mm} \rightarrow d_i = 200\text{mm}\ (2f : 2f)$

$d_o = 150\text{mm} \rightarrow d_i = 300\text{mm}$

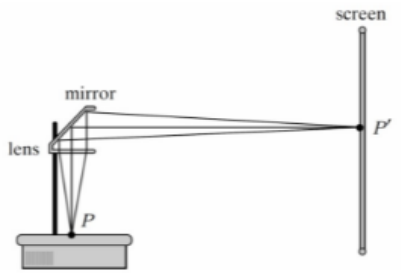
$d_o = 50\text{mm} \rightarrow d_i = -100\text{mm}$



Projector

The lens in an overhead projector forms an image P' of a point P on an overhead transparency.

If the screen is moved closer to the projector, must the lens be:



1. moved up
2. left in place
3. moved down

to keep the image on the screen in focus?

1. Moved up

2. Left in place

3. Moved down

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GAMSAT style question on lenses

Questions 22 and 23

A thin luminous rectangular plate, 2 cm × 3 cm, faces a thin converging lens of focal length 20 cm. The plate is perpendicular to the principal axis of the lens and its centre lies on this axis. When the plate is placed at certain distances from the lens, real rectangular images of the plate are formed on a moveable screen held perpendicular to the principal axis.

For thin lenses two useful formulae are:

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ and $M = \frac{h}{H} = \frac{|v|}{|u|}$

Where: f = focal length of the lens

u = distance from lens to object

v = distance from lens to image

M = magnitude of magnification

H = height of object

h = height of image

GAMSAT style question continued…

22. If the plate is positioned 30 cm from the lens, what is the distance of its real image from the lens?

A. 10 cm

B. 20 cm

C. 30 cm

D. 60 cm

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \Rightarrow \mathrm{v} = \left(\frac{1}{20} - \frac{1}{30}\right)^{-1} = (0.05 - 0.0333)^{-1} = 60\ cm$$

23. If the plate is positioned 30 cm from the lens, what are the dimensions of the rectangular image formed on the screen?

A. 1 cm x 1.5 cm

B. 2 cm x 3 cm

C. 4 cm x 6 cm

D. 8 cm x 12 cm

$M = \frac{|v|}{|u|} = \frac{60}{30} = 2$ so plate dimensions are increased by a factor of two.

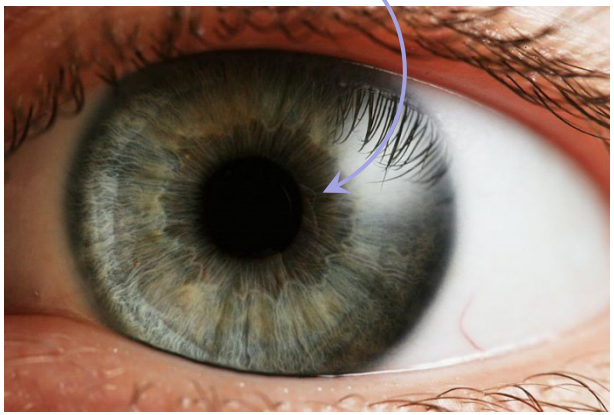
(4 cm x 6 cm)

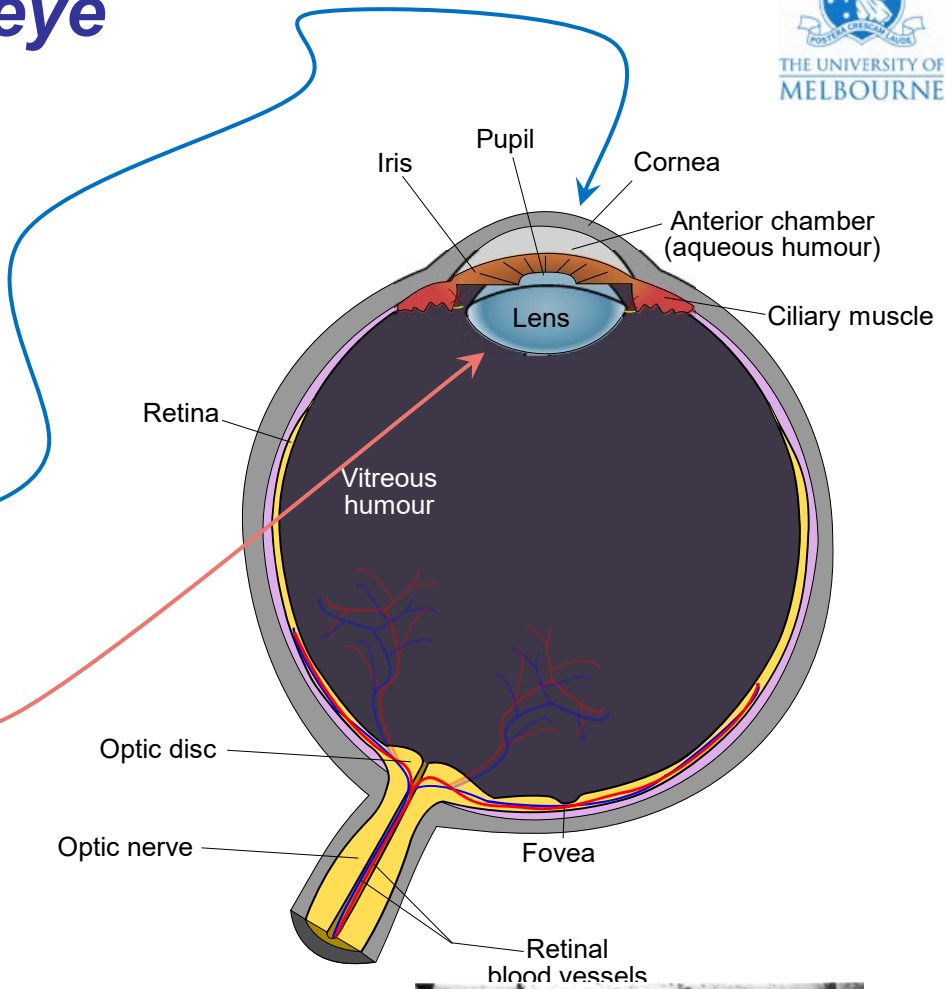
Chapter 25

OPTICAL SYSTEMS: THE EYE

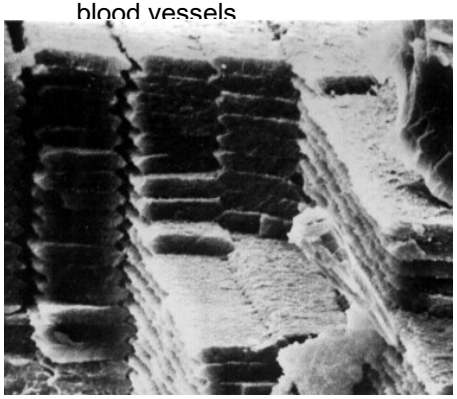
25.2 Optical systems: the eye

- The eye: lens, high resolution colour sensitive imaging detector, data communications, control mechanisms, protection & self-repair mechanisms, etc.
- Compound optical system: positive double-lens + spherical light-sensitive surface
- Most of focusing at air-cornea interface
- *Accommodation* (focal length variation) by crystalline lens
- Aperture stop (iris-pupil) controls irradiance and aberrations





Lens material: 22,000 layers continually growing (until age ~ 48 y.o.)



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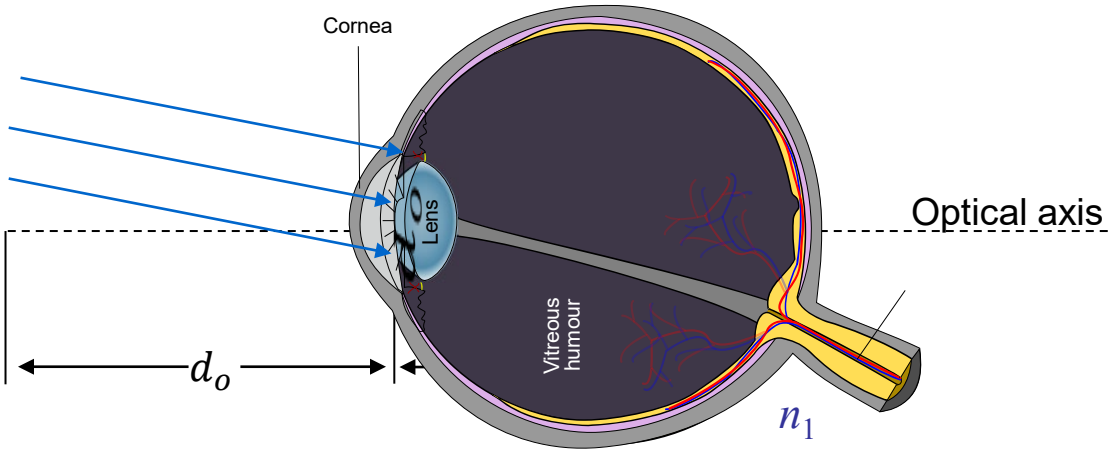
The eye: two lenses



The human eye is a roughly spherical lens of diameter 25mm. It focuses incoming parallel rays on the retina. Two focusing components:

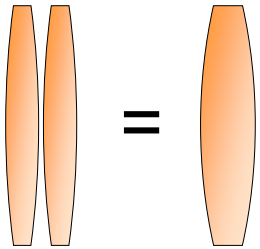
- Roughly spherical ball of *vitreous humour* (i.e. the goo inside the eyeball) of diameter 25mm. Refractive index of *vitreous humour n* = 1.336
- Focus primarily defined by smaller (~ 8mm) radius of cornea
- Additional crystalline lens, similar refractive index but much smaller radius of curvature
- Radius of curvature of "lens" controlled by muscles

FOCAL LENGTH AS LENS POWER

Lens power

- We have $\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$

- Two thin lenses <u>in contact</u>: $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$



- Often convenient to use "lens power": $P = \frac{1}{f}$
  - If f is in metres, then power is in ***diopters:*** +1 diopter ≡ 1 dpt ≡ 1.0 D ≡ 1 m^{-1}
  - Commonly used in optometry, e.g. +1 diopter (moderate), +3 diopters (strong)
  - Negative if diverging lens

- Two thin lenses in contact, just add "powers": $P = P_1 + P_2$

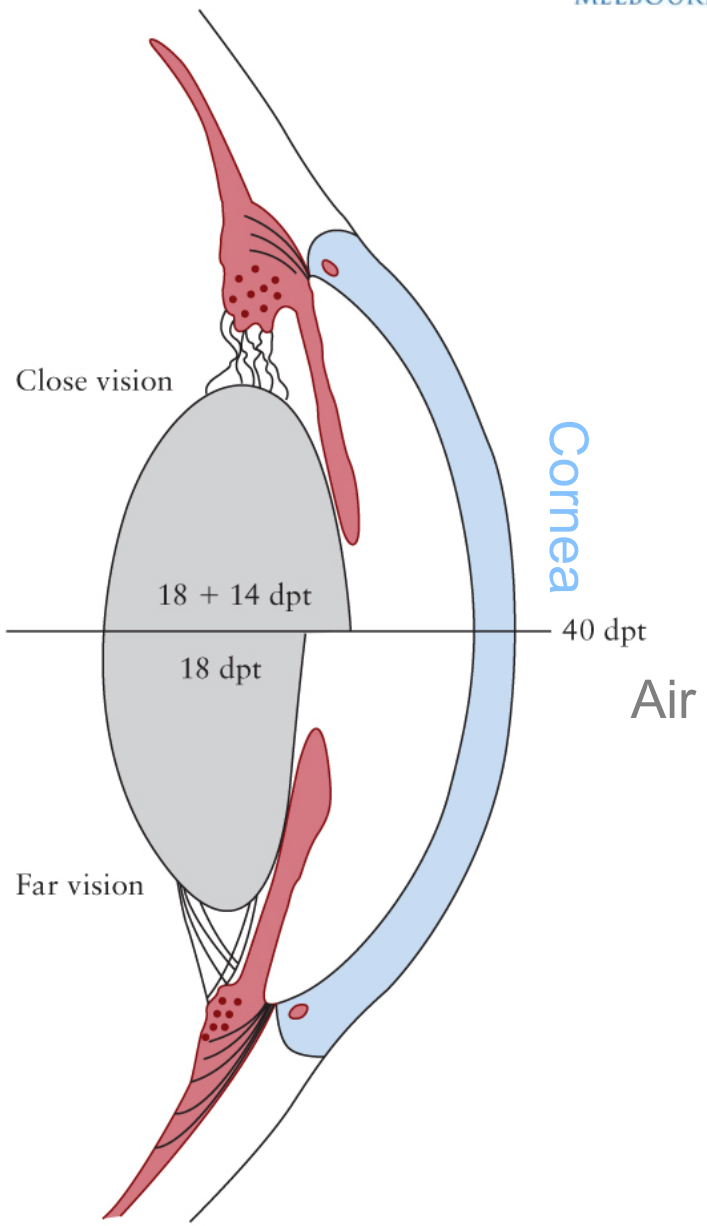
Two lenses: cornea/air + "lens"

- Cornea/air refracts
  - Radius ≈ 8mm
  - Refractive index air = 1, goo = 1.33
  - $P_{\text{cornea}} = \frac{\Delta n}{R} = 40$ dpt

- "Lens" refracts ($n \sim 1.4$)
  - Ciliary muscles change shape of "lens"
  - $P_{\text{lens}} = 18$ dpt to 32 dpt

- Combined:
  - $P_{\text{eye}} = P_{\text{cornea}} + P_{\text{lens}}$
  - $\mathfrak{N}_{\text{eye}} = 40 + 18 \text{ to } 32 = 58 \text{ to } 72$ dpt

- Focal length $f = \frac{1}{P_{\text{eye}}} = 14$mm to 17mm

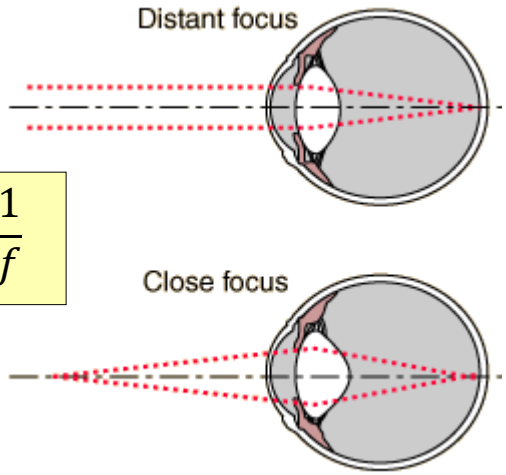


THE EYE: VISION DEFECTS

Focussing and accommodation

- The Lens has only small focussing effect (small difference in n)
- Most focussing at cornea (air/goo interface)
- Lens = fine-tuning: i.e. “focus”, accommodation
- Ciliary muscle: *changes radius of curvature*
  - Relaxed → focus at infinity (beyond approx 5m)
  - Contract → shorter focal length → focus on closer objects

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$$



- Fish move lens back and forth,
  Mollusks expand/contract whole eye

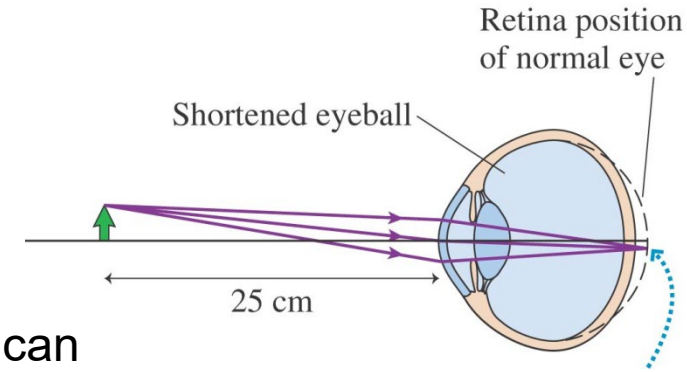
- Changing focal length when changing view = *accommodation*

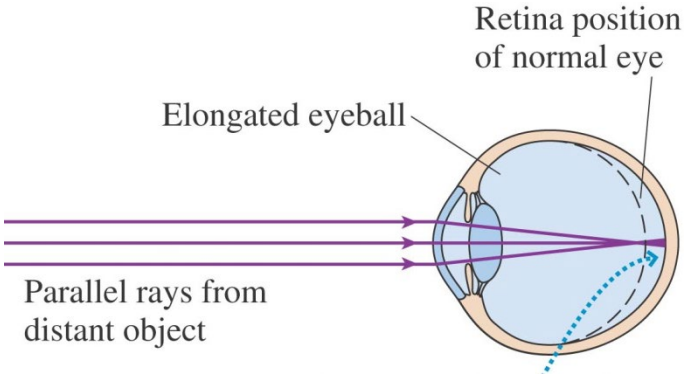
Vision defects and correction: $\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$

- Presbyopia: inability to accommodate (age-related)

- Hyperopia: far-sighted (age-related)
  - Eyeball too short (focal length too long)
  - Need extra lens power (converging)
  - Near point NP: closest distance at which eye can focus; normally 25cm but for far-sighted,

    $NP_{farsighted} > 25\text{cm}$



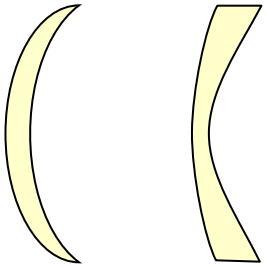


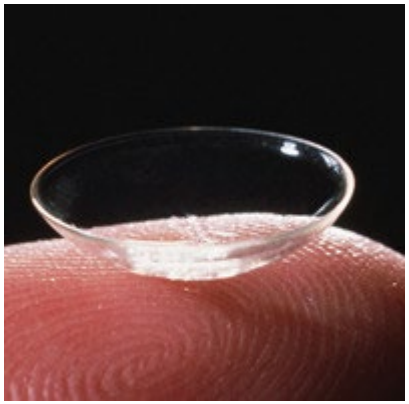
- Myopia: near-sighted
  - Eyeball too long (focal length too short)
  - Rays from distant object focus too soon
  - Need less lens power (diverging)
  - Far point FP: distance at which completely relaxed eye can focus; normally infinity but for near-sighted

    $FP_{nearsighted} < \infty$

Vision defects and correction: $\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$

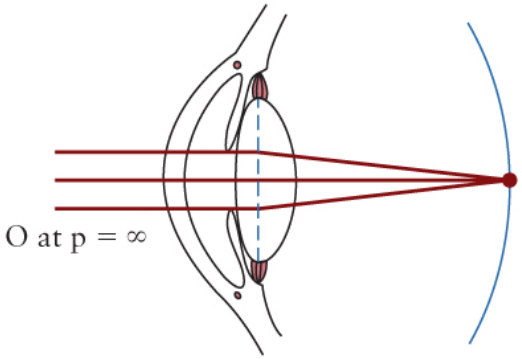
- Easily corrected: eyeglasses, contact lenses, laser keratotomy
- Required correction generally measured as ***lens power:*** $P = \frac{1}{f}$
  - in ***diopters:*** +1 diopter ≡ 1 dpt ≡ 1.0 D ≡ 1 m^{-1}



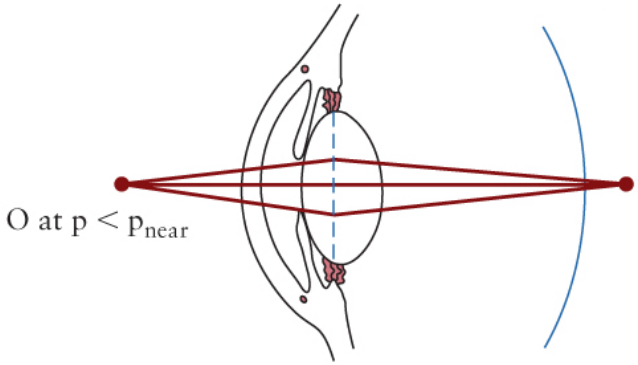


Farsighted (hyperopia)

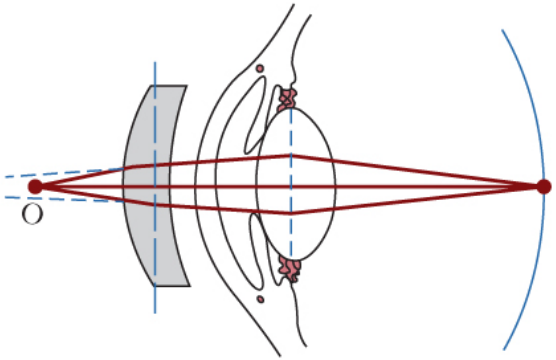
- Object far away: all good



- Object close
  - eye focus not strong enough

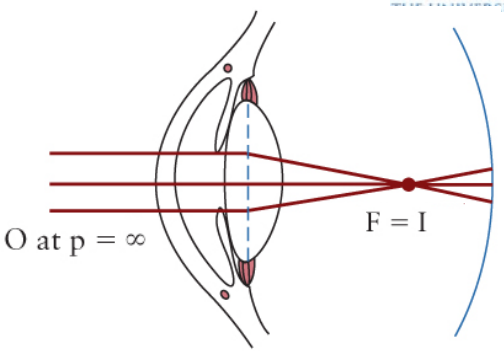


- Increase power: add converging lens

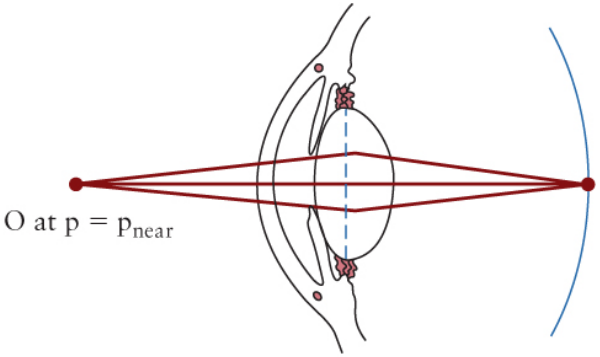


Nearsighted (myopia)

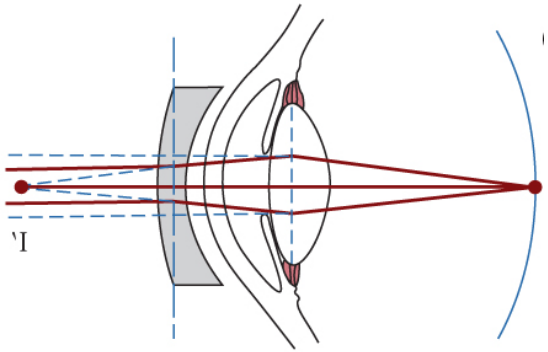
- Object far away: eye focus too strong



- Object close: all good



- Decrease power: add diverging lens



D Oa-2 Model of eye

The eye: summary

- Cornea, crystalline lens, ball lens, pupil, retina

- Accommodation: focusing

- Vision defects
  - myopia, near-sighted, lens too strong
  - hyperopia, far-sighted, lens too weak

- Corrective lenses:
  - myopia, need diverging lens
  - hyperopia, need converging lens

- Lens power $P = \frac{1}{f}$ (diopters, dpt, D, m^{-1})

Summary: Geometric Optics

Geometric optics: light as rays

- Lens transforms angle to position $y = -f\,\theta$
- Thin lens equation $$\frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}$$
- Imaging: for each point, draw three "special" rays:
  1. Through optic centre O
  2. Parallel to optic axis, then through far focal point F_i
  3. Through near focal point F_o then parallel to optic axis
- The Eye
  - Air/cornea + lens
  - Far-sighted, lens too weak (hyperopia)
  - Near-sighted, lens too strong (myopia)

That's all from me…

No Lectures, Labs or Problem Solving classes during the mid-semester break next week.

Professor Phillip Urquijo will be your Lecturer starting the week of April 17th.

Thank you!